

Homework problemsSOLUTIONS

① (a) $f(x) = x$, $-\pi < x < \pi$

This is an odd function, hence $a_n = 0$ and only sine terms contribute to the Fourier series (FS).

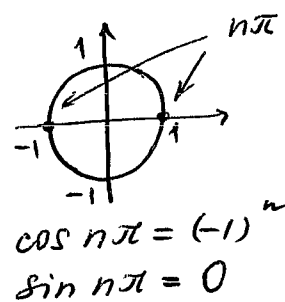
$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} \underbrace{x \sin nx}_{\text{even integrand}} dx = \frac{2}{\pi} \int_0^{\pi} x \sin nx dx$$

$$= \frac{2}{\pi} \left[x \frac{1}{n} (-\cos nx) \right]_0^{\pi} + \frac{1}{n} \int_0^{\pi} \cos nx dx$$

$$= \frac{2}{\pi} \left[\frac{\pi}{n} \underbrace{(-\cos n\pi)}_{(-1)^n} + \frac{1}{n^2} \underbrace{\sin nx}_0 \right]_0^{\pi}$$

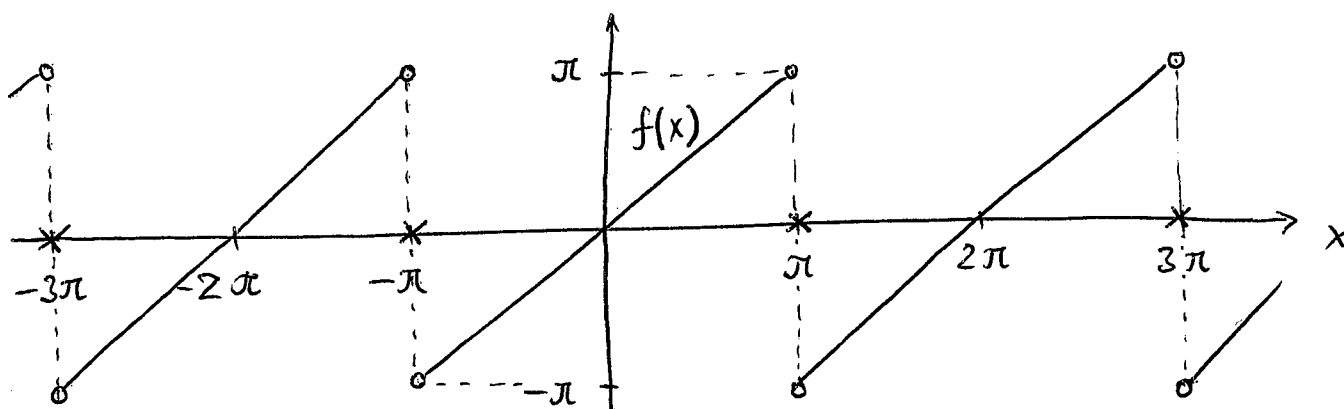
$$= \frac{2}{n} (-1)^{n-1}$$

Hence, $FS = 2 \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} \sin nx$.



This Fourier series gives a periodic extension of $f(x)$ shown on this graph:

x - values at discontinuities



Particular value: $x = \frac{\pi}{2}$, $f(x) = \frac{\pi}{2}$

$$\frac{\pi}{2} = 2 \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} \sin \frac{n\pi}{2} \Rightarrow \frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$$

$\begin{matrix} (-1)^m, & n=2m+1 \text{ (odd)} \\ 0, & n=2m \text{ (even)} \end{matrix}$

(b) $f(x) = x^2, -\pi \leq x \leq \pi$

(2)

Here $f(x)$ is even, hence all $b_n = 0$, and the Fourier series contains only cosine terms (and possibly a_0).

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 \cos nx \, dx = \frac{2}{\pi} \int_0^{\pi} x^2 \cos nx \, dx$$

$$n = 0 : \quad a_0 = \frac{2}{\pi} \int_0^{\pi} x^2 \, dx = \frac{2}{\pi} \frac{x^3}{3} = \frac{2\pi^2}{3}$$

$$n > 0 : \quad a_n = \frac{2}{\pi} \left[\underbrace{x^2 \frac{1}{n} \sin nx}_{=0} \Big|_0^{\pi} - \frac{1}{n} \int_0^{\pi} \sin nx \cdot 2x \, dx \right]$$

(integrating by parts)

$$= \frac{2}{\pi} \left[-\frac{2}{n^2} x (-\cos nx) \Big|_0^{\pi} - \frac{2}{n^2} \int_0^{\pi} \cos nx \, dx \right]$$

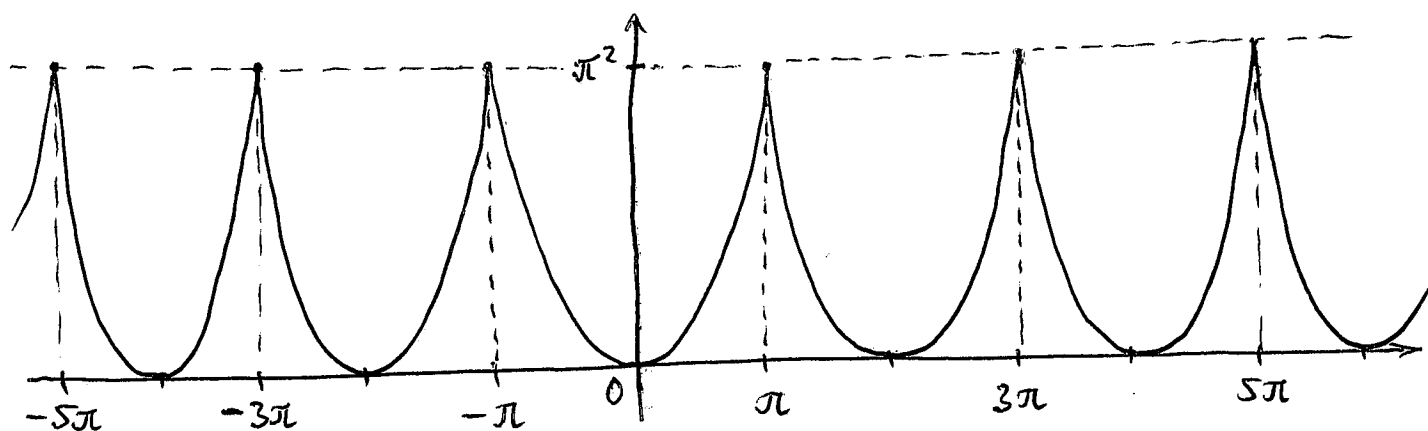
$$= \frac{2}{\pi} \left[\frac{2\pi}{n^2} \underbrace{\cos n\pi}_{(-1)^n} - \frac{2}{n^3} \underbrace{\sin nx}_{=0} \Big|_0^{\pi} \right]$$

$$= \frac{4(-1)^n}{n^2}$$

Hence, the FS is

$$\frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} (-1)^n \frac{\cos nx}{n^2}$$

This Fourier series represents a periodic extension of $f(x) = x^2, -\pi \leq x \leq \pi$:

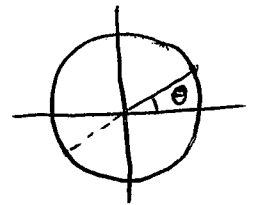


The extension has no discontinuities.

(c) $f(x) = \sin \alpha x$, $-\pi < x < \pi$, $\alpha \notin \mathbb{Z}$. (3)

This is an odd function, so $a_n = 0$, and

$$\begin{aligned}
 b_n &= \frac{2}{\pi} \int_0^{\pi} \sin \alpha x \sin nx \, dx \\
 &= \frac{2}{\pi} \int_0^{\pi} \frac{1}{2} [\cos(\alpha-n)x - \cos(\alpha+n)x] \, dx \\
 &= \frac{1}{\pi} \left[\frac{1}{\alpha-n} \sin(\alpha-n)x - \frac{1}{\alpha+n} \sin(\alpha+n)x \right]_0^{\pi} \\
 &= \frac{1}{\pi} \left[\frac{\sin(\alpha-n)\pi}{\alpha-n} - \frac{\sin(\alpha+n)\pi}{\alpha+n} \right] \left\{ \begin{array}{l} \sin(\alpha\pi + n\pi) \\ = \sin \alpha\pi (-1)^n \end{array} \right. \\
 &= \frac{1}{\pi} (-1)^n \sin \alpha\pi \left[\frac{1}{\alpha-n} - \frac{1}{\alpha+n} \right] \\
 &= \frac{1}{\pi} (-1)^n \sin \alpha\pi \frac{\alpha+n - \alpha+n}{\alpha^2 - n^2} \\
 &= \frac{2 \sin \alpha\pi}{\pi} (-1)^{n-1} \frac{n}{n^2 - \alpha^2}
 \end{aligned}$$



$$\begin{aligned}
 \sin(\theta + n\pi) \\
 = (-1)^n \sin \theta
 \end{aligned}$$

Fourier series: $\sin \alpha x \rightarrow \frac{2 \sin \alpha\pi}{\pi} \sum_{n=1}^{\infty} (-1)^{n-1} \frac{n \sin nx}{n^2 - \alpha^2}$

(2) From Q.1(b)

$$x^2 = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} (-1)^n \frac{\cos nx}{n^2}, \quad -\pi \leq x \leq \pi$$

Using $x=0$: $0 = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2} = \frac{\pi^2}{12},$$

i.e.

$$1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots = \frac{\pi^2}{12}$$

Using $x=\pi$:

$$\pi^2 = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} (-1)^n \frac{\cos n\pi}{n^2}$$

($f(\pi) = f(\pi)$)

$$\cos n\pi = (-1)^n$$

$$\frac{2}{3} \pi^2 = 4 \sum_{n=1}^{\infty} \frac{1}{n^2} \Rightarrow \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6},$$

i.e. $1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots = \frac{\pi^2}{6}.$

③ $f(x) = |\sin x|, \quad -\pi \leq x \leq \pi$

This is an even function, so $b_n = 0$, and

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} |\sin x| \cos nx \, dx = \frac{2}{\pi} \int_0^{\pi} \sin x \cos nx \, dx$$

($\sin x \geq 0$ on $0 \leq x \leq \pi$, so the modulus sign can be removed).

$n=0:$ $a_0 = \frac{2}{\pi} \int_0^{\pi} \sin x \, dx = \frac{2}{\pi} (-\cos x) \Big|_0^{\pi}$

$$= \frac{2}{\pi} \left(\underbrace{-\cos \pi}_{-1} + \underbrace{\cos 0}_1 \right) = \frac{4}{\pi}$$

$n > 0:$ $a_n = \frac{2}{\pi} \int_0^{\pi} \frac{1}{2} [\sin(1+n)x + \sin(1-n)x] \, dx$

$$= \frac{1}{\pi} \left[\frac{1}{1+n} (-\cos(1+n)x) \Big|_0^{\pi} + \frac{1}{1-n} (-\cos(1-n)x) \Big|_0^{\pi} \right]$$

$$= \frac{1}{\pi} \left[\frac{1}{1+n} \left(1 - \underbrace{\cos(n+1)\pi}_{\substack{(-1)^{n+1} \\ = (-1)^{n-1}}} \right) + \frac{1}{1-n} \left(1 - \underbrace{\cos(n-1)\pi}_{\substack{(-1)^{n-1} \\ \left(\cos(1-n)x = \cos(n-1)\pi \text{ even function} \right)}} \right) \right]$$

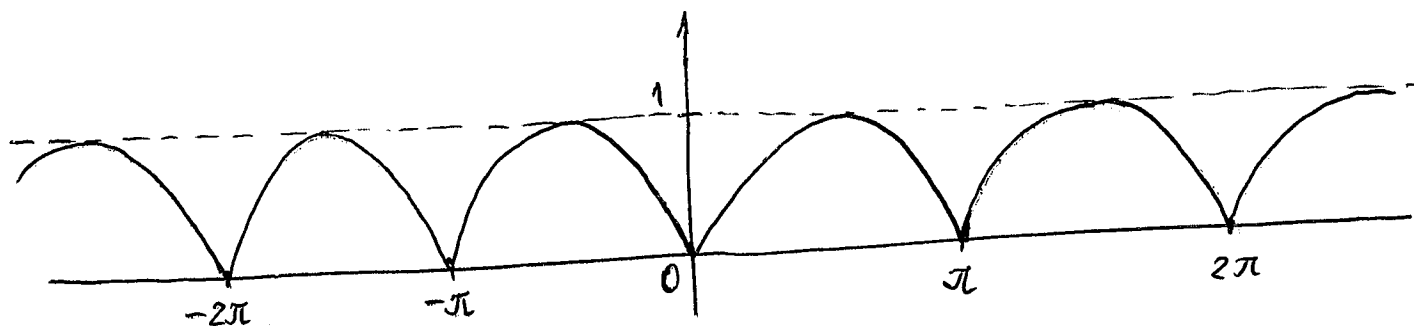
$$= \frac{1}{\pi} \underbrace{\left(1 - (-1)^{n-1} \right)}_{\substack{0 \text{ for odd } n \\ 2 \text{ for even } n}} \cdot \underbrace{\left[\frac{1}{1+n} + \frac{1}{1-n} \right]}_{\frac{2}{1-n^2}} = \begin{cases} 0, & \text{odd } n \\ -\frac{4}{\pi} \frac{1}{n^2-1}, & n = 2m \text{ (even } n) \end{cases}$$

$m = 1, 2, \dots$

Hence, the Fourier series contains only even terms with $n = 2m$, and we have:

$$|\sin x| = \underbrace{\frac{2}{\pi}}_{\frac{a_0}{2}} - \frac{4}{\pi} \sum_{m=1}^{\infty} \frac{\cos 2mx}{4m^2 - 1}, \text{ as required.} \quad (5)$$

This Fourier series describes a periodic extension of $|\sin x|$:



Taking $x=0$,

$$0 = \frac{2}{\pi} - \frac{4}{\pi} \sum_{m=1}^{\infty} \frac{\cos 0}{4m^2 - 1}$$

$$\frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{4n^2 - 1} = \frac{2}{\pi}$$

(we have replaced m by n here)

$$\sum_{n=1}^{\infty} \frac{1}{4n^2 - 1} = \frac{1}{2}$$

In fact, it is easy to show, e.g., by induction that

$$\sum_{n=1}^N \frac{1}{4n^2 - 1} = \frac{N}{2N+1} \xrightarrow{N \rightarrow \infty} \frac{1}{2}$$

Or, even simpler, by noticing:

$$\sum_{n=1}^{\infty} \frac{1}{4n^2 - 1} = \sum_{n=1}^{\infty} \frac{1}{(2n-1)(2n+1)} = \sum_{n=1}^{\infty} \frac{1}{2} \left[\frac{1}{2n-1} - \frac{1}{2n+1} \right]$$

$$= \frac{1}{2} \left[\left(1 - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{5}\right) + \left(\frac{1}{5} - \frac{1}{7}\right) + \dots \right]$$

$$= \frac{1}{2}$$

(4)

6

$$f(x) = \begin{cases} 0, & -\pi \leq x \leq 0 \\ \sin x, & 0 \leq x \leq \pi \end{cases}$$

$$a_n = \frac{1}{\pi} \int_0^{\pi} \sin x \cos nx \, dx$$

The integral is from 0 to π ,
as $f(x) = 0$ on $-\pi \leq x \leq 0$

Using some results from Q.3, and noting that these are equal $\frac{1}{2}$ of those derived in Q.3, we have

$$a_0 = \frac{2}{\pi}, \quad a_n = -\frac{2}{\pi} \frac{1}{n^2-1}, \quad n = 2m$$

$$a_n = 0, \quad n = 2m-1$$

$$b_n = \frac{1}{\pi} \int_0^{\pi} \sin x \sin nx \, dx$$

$$\underline{n=1}: \quad b_1 = \frac{1}{\pi} \int_0^{\pi} \sin^2 x \, dx$$

$$\cos 2x = 1 - 2\sin^2 x$$

$$\sin^2 x = \frac{1 - \cos 2x}{2}$$

$$= \frac{1}{\pi} \cdot \frac{1}{2} \int_0^{\pi} (1 - \cos 2x) \, dx$$

$$= \frac{1}{2\pi} \left[\underbrace{x \Big|_0^{\pi}}_{\pi} - \frac{1}{2} \underbrace{\sin 2x \Big|_0^{\pi}}_0 \right] = \frac{1}{2}$$

$$\underline{n \neq 1}: \quad b_n = \frac{1}{\pi} \int_0^{\pi} \frac{1}{2} [\cos (1-n)x - \cos (1+n)x] \, dx$$

$$= \frac{1}{2\pi} \left[\frac{1}{1-n} \sin(1-n)x \Big|_0^{\pi} - \frac{1}{1+n} \sin(1+n)x \Big|_0^{\pi} \right]$$

$$= 0$$

Hence, we have the Fourier series:

$$\frac{1}{\pi} - \frac{2}{\pi} \sum_{m=1}^{\infty} \frac{\cos 2mx}{4m^2-1} + \frac{1}{2} \sin x$$

This is $\frac{1}{2}$ of the Fourier series from Q.3.

This result could have been written straightaway, (7) by noticing that in this question

$$f(x) = \frac{1}{2} |\sin x| + \frac{1}{2} \sin x$$

Indeed: for $0 \leq x \leq \pi$ $\sin x \geq 0$, so

this gives $f(x) = \sin x$;

for $-\pi \leq x \leq 0$ $\sin x \leq 0$, so

this gives $f(x) = -\frac{1}{2} \sin x + \frac{1}{2} \sin x = 0$.

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For $x = -\frac{\pi}{2}$ we have: $f(x) = 0$, so that

$$0 = \frac{1}{\pi} - \frac{2}{\pi} \sum_{m=1}^{\infty} \frac{\cos 2m(-\frac{\pi}{2})}{4m^2 - 1} + \frac{1}{2} \underbrace{\sin(-\frac{\pi}{2})}_{=-1}$$

$$\cos(-m\pi) = (-1)^m$$

$$\frac{2}{\pi} \sum_{m=1}^{\infty} \frac{(-1)^m}{4m^2 - 1} = \frac{1}{\pi} - \frac{1}{2}$$

$$\sum_{m=1}^{\infty} \frac{(-1)^{m-1}}{4m^2 - 1} = \frac{1}{2} - \frac{\pi}{4}.$$

[Note: we could also obtain this result by setting $x = \frac{\pi}{2}$, or by setting $x = \frac{\pi}{2}$ in the Fourier series from Q.3.]

(5) $f(x) = x(\pi - x), \quad 0 \leq x \leq \pi$ (8)

Expanding this function in a sine Fourier series, we obtain its odd extension.

$$b_n = \frac{2}{\pi} \int_0^{\pi} x(\pi - x) \sin nx \, dx$$

$$= \frac{2}{\pi} \left[\pi \int_0^{\pi} x \sin nx \, dx - \int_0^{\pi} x^2 \sin nx \, dx \right]$$

$$= 2 \underbrace{\int_0^{\pi} x \sin nx \, dx}_I - \frac{2}{\pi} \underbrace{\int_0^{\pi} x^2 \sin nx \, dx}_{II}$$

Integral I: $\int_0^{\pi} x \sin nx \, dx = x \frac{1}{n} (-\cos nx) \Big|_0^{\pi} + \frac{1}{n} \int_0^{\pi} \cos nx \, dx$

$$= -\frac{\pi}{n} \underbrace{\cos n\pi}_{(-1)^n} + \frac{1}{n^2} \sin nx \Big|_0^{\pi} = -(-1)^n \frac{\pi}{n}$$

Integral II: $\int_0^{\pi} x^2 \sin nx \, dx = x^2 \frac{1}{n} (-\cos nx) \Big|_0^{\pi} + \frac{2}{n} \int_0^{\pi} \cos nx \, x \, dx$

$$= \frac{\pi^2}{n} \underbrace{(-\cos n\pi)}_{(-1)^n} + \underbrace{x \frac{2}{n^2} \sin nx \Big|_0^{\pi}}_0 - \frac{2}{n^2} \int_0^{\pi} \sin nx \, dx$$

$$= -(-1)^n \frac{\pi^2}{n} + \frac{2}{n^3} \cos nx \Big|_0^{\pi}$$

$$= -(-1)^n \frac{\pi^2}{n} + \frac{2}{n^3} (\cos n\pi - 1) = -(-1)^n \frac{\pi^2}{n} - \frac{2}{n^3} (1 - (-1)^n)$$

So, we have:

$$b_n = \cancel{-2(-1)^n \frac{\pi}{n}} + \cancel{\frac{2}{\pi}(-1)^n \frac{\pi^2}{n}} + \frac{2}{\pi} \frac{2}{n^3} (1 - (-1)^n)$$

$$= \frac{4}{\pi} \frac{1}{n^3} (1 - (-1)^n) = \begin{cases} 0, & n \text{ even} \\ \frac{8}{\pi} \frac{1}{n^3}, & n = 2m+1 \text{ (odd)} \end{cases}$$

Hence, the sine Fourier series for $f(x)$ is:

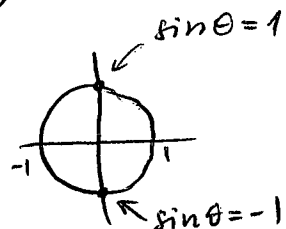
(9)

$$\frac{8}{\pi} \sum_{m=0}^{\infty} \frac{\sin(2m+1)x}{(2m+1)^3} \quad (*)$$

For $x = \frac{\pi}{2}$ $x(\pi-x) = \frac{\pi}{2} \cdot \frac{\pi}{2} = \frac{\pi^2}{4}$

In the series $\sin(2m+1)\frac{\pi}{2} = \sin(m+\frac{1}{2})\pi$
 $= (-1)^m$

Hence: $\frac{8}{\pi} \sum_{m=0}^{\infty} \frac{(-1)^m}{(2m+1)^3} = \frac{\pi^2}{4}$



$$\sum_{m=0}^{\infty} \frac{(-1)^m}{(2m+1)^3} = \frac{\pi^3}{32}$$

It is interesting to note that the Fourier series (*),

$$\frac{8}{\pi} \left[\sin x + \frac{\sin 3x}{27} + \frac{\sin 5x}{125} + \dots \right]$$

has a very large relative contribution of the 1st term. This is a result of the odd-extension of $f(x) = x(\pi-x)$, $0 \leq x \leq \pi$, being very similar to the $\sin x$ function:

